

Nervous systems consist of circuits of neurons and supporting cells

The ability to sense and react originated billions of years ago in prokaryotes, enhancing survival and reproductive success in changing environments. Later in evolution, modification of simple recognition and response processes provided a basis for communication between cells in an animal body. By the time of the Cambrian explosion, more than 500 million years ago (see Concept 32.2), specialized nervous systems had appeared that enable animals to sense their surroundings and respond rapidly.

Hydras, jellies, and other cnidarians are the simplest animals with nervous systems. In most cnidarians, interconnected neurons form a diffuse *nerve net* (**Figure 49.2a**), which controls the contraction and expansion of the gastrovascular cavity. In more complex animals, the axons of multiple neurons are often bundled together, forming **nerves**. These fibrous structures channel information flow along specific routes through the nervous system. For example, sea stars have a set of radial nerves connecting to a central nerve ring (**Figure 49.2b**). Within each arm of a sea star, the radial nerve is linked to a nerve net from which it receives

input and to which it sends signals that control muscle contraction.

Animals with elongated, bilaterally symmetrical bodies have even more specialized nervous systems. In particular, they exhibit *cephalization*, an evolutionary trend toward a clustering of sensory neurons and interneurons at the anterior (front) end of the body. Nerves that extend toward the posterior (rear) end enable these anterior neurons to communicate with cells elsewhere in the body.

In many animals, neurons that carry out integration form a **central nervous system (CNS)**, and neurons that carry information into and out of the CNS form a **peripheral nervous system (PNS)**. In nonsegmented worms, such as the planarian in **Figure 49.2c**, a small brain and longitudinal nerve cords constitute the simplest clearly defined CNS. In certain nonsegmented worms, the entire nervous system is constructed from only a small number of cells, as in the case of the nematode *Caenorhabditis elegans*. In this species, an adult worm (hermaphrodite) has exactly 302 neurons, no more and no fewer. More complex invertebrates, such as segmented worms (**Figure 49.2d**) and arthropods (**Figure 49.2e**), have many more neurons. Their behavior is regulated by more complicated brains and by ventral nerve cords containing **ganglia**, segmentally arranged clusters of neurons that act as relay points in transmitting information.

▼ **Figure 49.2 Nervous system organization.** (a) A hydra contains individual neurons (purple) organized in a diffuse nerve net. (b–h) Animals with more sophisticated nervous systems contain groups of neurons (blue) organized into nerves and often ganglia and a brain.

